

New Zealand Journal of Geology and Geophysics

ISSN: 0028-8306 (Print) 1175-8791 (Online) Journal homepage: www.tandfonline.com/journals/tnzg20

The paleomagnetism of some rock samples from Viti Levu, Fiji

D. H. Tarling

To cite this article: D. H. Tarling (1967) The paleomagnetism of some rock samples from Viti Levu, Fiji, New Zealand Journal of Geology and Geophysics, 10:5, 1235-1247, DOI: 10.1080/00288306.1967.10420215

To link to this article: https://doi.org/10.1080/00288306.1967.10420215

Published online: 21 Dec 2011.

Submit your article to this journal

Article views: 243

View related articles

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tnzg20

THE PALEOMAGNETISM OF SOME ROCK SAMPLES FROM VITI LEVU, FIJI

D. H. TARLING

Department of Geophysics and Planetary Physics, School of Physics, the University of Newcastle upon Tyne, England

(Received for publication 11 January 1967)

ABSTRACT

The natural remanence of 126 samples of igneous and sedimentary rock is studied. The remanence of the sedimentary rocks is found unsuitable for interpreting in terms of ancient geomagnetic field directions. Samples of igneous rock are found to be of varying stability but clearly merit further study, if more samples could be obtained, as such data would be of direct relevance to stratigraphic correlation and the tectonic history of this region.

INTRODUCTION

The Crown Colony of Fiji comprises about 300 islands, of which the two largest are Viti Levu and Vanua Levu. Viti Levu forms over half the total area of the Colony (10,500 km^2 of 18,250 km^2) and is mountainous, rising over 1,000 m in many places. The island was visited in 1960, and 126 oriented samples were obtained from 58 sites (Fig. 1), with the assistance of the Fiji Geological Survey. This was a reconnaissance survey forming a small part of a paleomagnetic study of late Cenozoic igneous rocks in and around the Pacific Basin (Tarling, 1963). The results presented here, therefore, are not intended as a definitive study, but may serve to outline the suitability of certain Fiji formations for future paleomagnetic study.

The intensity and direction of remanence were determined for specimens, which were drilled from each sample, using astatic magnetometers (Blackett, 1952; Collinson *et al.*, 1957). Viscous components of the natural remanence were then removed or reduced in all specimens by partial demagnetisation in 150 (peak) oersteds (Creer, 1958), and the remanence remeasured. The stability of remanence to increasing temperature or alternating field strength was tested on a few pilot specimens by partial step demagnetisation (Irving *et al.*, 1961a, 1961b), and the laboratory stability index determined (Tarling and Symons, 1967) using three or more successive observations of the remanent vector over an increasing range of temperature or field (Stability Index = max (range$^{\frac{1}{2}}$ circular standard deviation^{-1})).

PALEOMAGNETIC RESULTS

The results are presented in stratigraphical sequence commencing with the oldest Group. The geology of each formation is described very briefly, as it

is discussed in further detail by Rodda and Band (1967), who give further references.

(i) The *Wainimala Group* is the oldest on Viti Levu, comprising a mixture of rock types including lavas, grits, breccias, argillites, and limestone. One of the limestones contains Upper Eocene fossils, and the Group extends into the Miocene. Many of the rocks have been altered, faulted, and folded by subsequent tectonic movements and associated intrusions. The 22 sites sampled in the Group (Table 1) were mostly sediments, but many appear to be either weathered or slightly metamorphosed. The scatter of initial remanence directions was large, and was only slightly improved after treatment in 150 (peak) oersteds, when most of the specimen remanent directions were north-easterly (Fig. 2), the mean direction giving equal weight to specimens being 56°, −19° with a circular standard deviation of 36°. Only three sites (10, 11, and 20) have significant mean directions after partial demagnetisation. Two specimens from site 17 (38a and 39b) were tested for stability in a.c. fields (Fig. 3); both showed poor stability (Table 2).

(ii) *Tholo Plutonics* intrude the Wainimala Group, but form a complex group showing considerable variation in both composition and time. One granitoid sample (site 41) has been dated by McDougall (1963) as

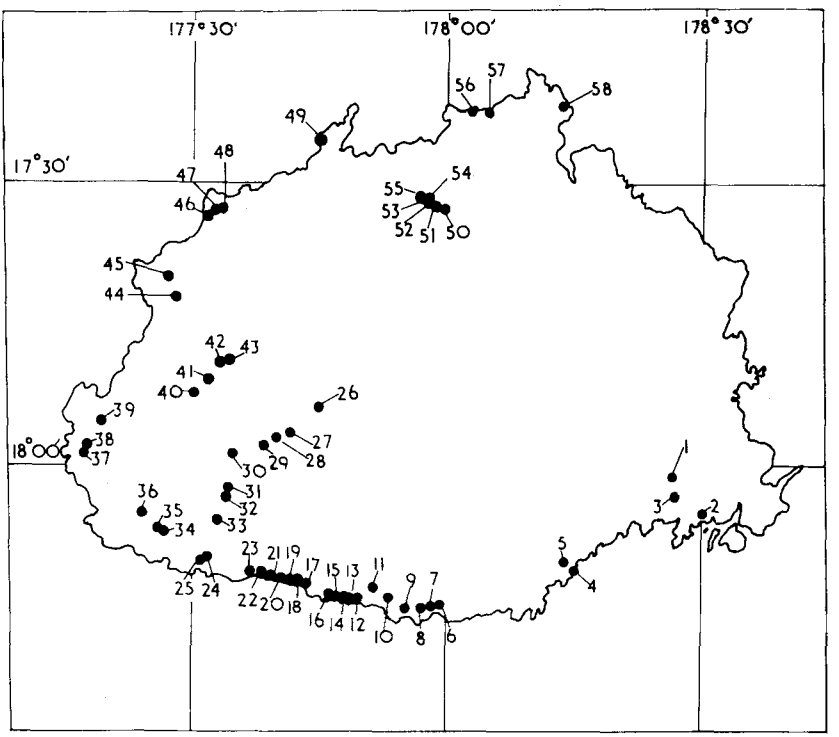


FIG. 1—Sampling sites on Viti Levu.

33 million years (m.y.). Four sites were sampled (Table 1), but the sample directions are widely scattered both before and after treatment in 150 (peak) oersteds; none of the sites which are testable has a significant mean direction. Two specimens (85a and 86a from site 41) were tested for remanence stability in a.c. fields (Fig. 3); one showed stability and the other poor stability (Table 2).

(iii) *Singatoka Sedimentary Group* comprises a mixture of argillites, sandstones, breccia, volcanics, and limestones which accumulated around the edges of the Wainimala Group and within fault troughs. Some of the limestones contain fossils of middle–upper Miocene age. The Group was sampled at five sites (Table 1). The initial remanent directions were somewhat scattered (Fig. 2), although the scatter was slightly reduced after treatment in 150 (peak) oersteds, and only two sites (26 and 30) have retained significant mean directions. Two specimens (51a and 52a from site 26) were tested for stability of remanence in a.c. fields (Fig. 3); one showed stability and the other poor stability (Table 2).

(iv) *Savura Volcanic Group* includes all volcanics in SE Viti Levu which lie stratigraphically between the Wainimala Group and the Suva Marl of the Mendrausuthu Group. Only one site was sampled (Table 1 and Fig. 2) in which the remanent directions were closely grouped both before and after partial demagnetisation in 150 (peak) oersteds). The mean direction was southerly, and of high negative inclination which departed strongly from both the Earth's present magnetic field and its reversed direction. To test if this deviation was attributable to anisotropy, one specimen (11b) was heated above its Curie point, and cooled in a magnetic field of 0·59 oersteds. The direction of thermoremanence acquired did not differ significantly from the applied field, so the deviation of the natural remanence cannot be attributed to anisotropy.

(v) *Mendrausuthu Andesitic Group* comprises mudstones, marls, limestones, tuffs, and andesites. Some limestones contain upper Miocene – lower Pliocene foraminifera and the Group is regarded as Mio-Pliocene in age. Seven sites were sampled (Table 1) in marls and sandstones, which showed varying degrees of weathering. The remanent directions, both before and after partial demagnetisation, were scattered (Fig. 2), although there appeared to be some clustering in a north-westerly direction, with streaking towards a south-easterly direction. The stability of remanence in alternating magnetic fields was tested on two specimens (Table 2 and Fig. 3); one showed stability (site 15) and the other poor stability (site 16). One site was sampled in an andesite at Mau (site 4). The sample directions were scattered initially, but became well grouped in a southerly direction after partial demagnetisation (Table 1 and Fig. 2).

(vi) *Nandi Sedimentary Group* comprises mudstones, marls, and tuffs, and was sampled in one site (Table 1 and Fig. 2). The single sample showed considerable change on partial demagnetisation, although agreement between the specimen directions remained good.

(vii) The *Koroimavua Andesitic Group* comprises volcanics and breccia overlying the Nandi Sedimentary Group, and overlain by parts of the Mba Basaltic Group. Only one site was sampled in the Lomolomo quarry, which

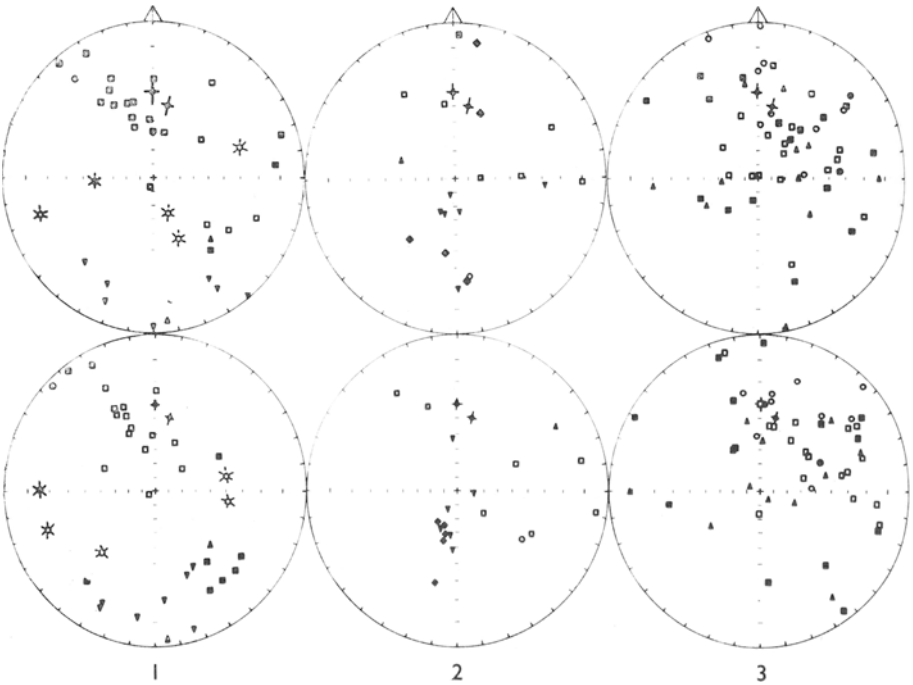


FIG. 2—Sample directions of remanence: initial (*top row*) and after treatment in 150 oersteds (*bottom row*). Crosses show the axial dipole and present Earth's magnetic field direction in Viti Levu. Solid symbols are of positive, downward inclination; hollow symbols are of negative, upward declination.

Column 1

○ Thuvu
△ Vanua Levu
□ Mba (Tavua)
inverted triangle Mba (Rakiraki)
star Nakombalevu

Column 2

○ Koroimavua
△ Nandi
□ Mendrausuthu (marls)
diamond Namosi
inverted triangle Savura

Column 3

○ Singatoka
△ Tholo
□ Wainimala

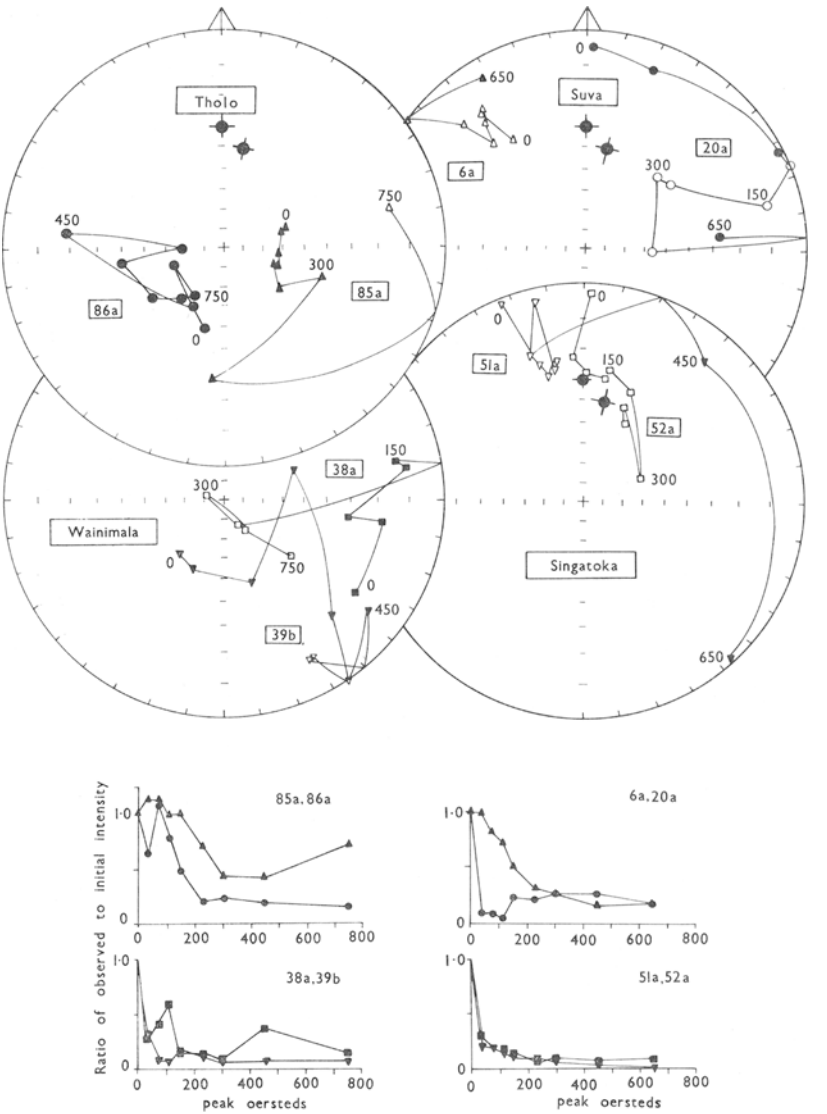


FIG. 3—Partial a.c. demagnetisation of pilot specimens from the Tholo Plutonics, Mendrausuthu (marls, including Suva Marl), Wainimala, and Singatoka Groups. (See also Table 2.) Solid and hollow symbols as in Fig. 2. Axial dipole and Earth's magnetic field direction also shown.

is the same site from which McDougall (1963) obtained samples which he subsequently dated by potassium/argon methods as 5·35 ± 0·1 m.y. The sample remanent direction was southerly and of low negative inclination, but became more easterly after partial demagnetisation (Table 1 and Fig. 2).

(viii) The *Nakombalevu Basalt* in SE Viti Levu is thought to be contemporaneous with the Mba Basaltic Group. The sample remanent directions (Table 1 and Fig. 2) were poorly grouped, both before and after partial demagnetisation.

(ix) The *Mba Basaltic Group* is mostly formed by olivine basalts erupted from several centres and overlying a strong Pliocene erosion surface. In some areas the lavas are intercalated with tuffs and marls containing upper Pliocene fossils. Fourteen lavas were sampled (Table 1 and Fig. 2), of which 11 belong to the Tavua volcano sites 45–55) and three (56–58) belong to the Rakiraki volcano. The exact stratigraphical relationship between the individual sites is not known, although the age probably increases from sites 45 to 49, which are probably lower than sites 50–55, in which sequence the stratigraphical order is reversed (i.e., the age decreases from site 55 to 50). One site (53) has been dated, using potassium/argon methods, as 4·3 ± 0·2 m.y. (N. H. Gale, pers. comm., 1966). However, there is some evidence that this site may have been reheated and this age must be considered to be a minimum estimate. The other three sites (56–58) are in lavas of the Rakiraki volcano, and are probably slightly younger than the other samples in the Mba Group.

The directions of remanence were variable. In site 47 the sample remanent directions were particularly widely scattered, but at most sites the mean vector was significant at a 95% level (Table 1) after partial demagnetisation. Sites 45 and 48–51 had remanent directions similar to the Earth's present magnetic field, but, although somewhat scattered, sites 46, 47, and 54–58 were reversed with respect to the present field. Sites 52 and 53 had intermediate remanent directions.

The stability of remanence in alternating magnetic fields was tested in sites 47, 49, and 57 (Fig. 4 and Table 2). Site 47 was found to be unstable; sites 49 and 57 were moderately stable. Samples from sites 51, 53, and 57 were tested for thermal stability of remanence (Table 2 and Fig. 4), and were all found to have poor to moderate stability.

Two specimens (116 from site 53, and 118 from site 54) were heated to 750°C, and cooled in a magnetic field of 0·59 oersteds. The acquired directions of remanence were the same as the applied field, suggesting an absence of anisotropic effects. Their acquired intensity of magnetisation was approximately 10 times that of their natural remanent magnetisation, suggesting substantial decay of their original remanence or, less likely, a much lower intensity of the Earth's magnetic field at the time the lavas cooled.

The coercivity of isothermal remanence was determined for seven specimens from sites 46, 47, 49, 53, 54, and 58. The values ranged from 130 oersteds (site 46) to 480 oersteds (site 49). This is consistent with the stability evidence in alternating magnetic fields, as it suggests that varying degrees of remanent stability can be expected in this Group.

TABLE 1—Mean Site Values of Remanence and Susceptibility

Site	N_{ss}	N_{sp}	Initial D (°)	Initial I (°)	Initial sig.	Initial csd (°)	Treated D (°)	Treated I (°)	Treated sig.	Treated csd (°)	$\bar{M}$ (gauss × 10^{-6})	$\bar{X}$ ($\frac{\text{gauss}}{\text{oersted}} \times 10^{-4}$)
Vanua Levu volcanics												
A	1	3	174	− 5	.....	.....	175	− 3	.....	.....	2486	.....
B	1	1	136	+33	.....	.....	134	+37	.....	.....	1473	.....
Viti Levu												
Thuvu Sedimentary Group[1]												
19	1	2	321	−12	.....	.....	315	+ 3	.....	.....	98	11
Mba Basaltic Group												
(a) Tavua volcano												
45	3	5	343	−18	0	18	341	−17	0	18	22032	3
46	2	4	78	+10	1	9	135	+19	2	10	685	27
47	3	6	26	−24	39	65	.....	.....	.....	.....	540	45
48	1	2	124	−28	.....	.....	345	−60	.....	.....	587	.....
49	3	6	347	−44	0	9	342	−47	0	9	1597	18
50	2	3	326	−12	7	22	327	−14	9	24	1994	92
51	3	6	332	−34	0	8	336	−33	0	4	918	95
52	1	2	194	−83	.....	.....	237	−84	.....	.....	38981	38
53[3]	2	4	13	−38	19	37	34	−60	2	13	301	52
54	2	4	135	− 6	32	49	147	+25	2	10	993	47
55	1	2	111	−18	.....	.....	134	+18	.....	.....	653	61
(b) Rakiraki volcano												
56	2	2	199	− 9	15	32	155	− 8	7	22	14	1
57	3	5	186	+ 6	4	34	200	+ 4	0	17	696	33
58	2	4	145	+ 8	3	15	156	+32	0	4	752	21

TABLE 1—*continued*

Site	N_{ss}	N_{sp}	Initial D (°)	Initial I (°)	Initial sig.	Initial csd (°)	Treated D (°)	Treated I (°)	Treated sig.	Treated csd (°)	$\bar{M}$ (gauss × 10^{-6})	$\bar{X}$ ($\frac{\text{gauss}}{\text{oersted}}$ × 10^{-4})
Nakombalevu Basalt												
3	5	9	193	−53	54	75	225	−52	46	72	634	31
Koroimavua Andesitic Group												
44[4]	1	2	172	−26	.....	.....	125	−35	.....	.....	237	65
Nandi Sedimentary Group[1]												
43	1	2	290	−49	.....	.....	57	−15	.....	.....	566	64
Mendrausuthu Andesitic Group												
4	5	10	187	+76	29	67	199	+53	0	15	392	35
1	1	3	329	−25	.....	.....	327	−17	.....	.....	100	1
5	1	1	2	+ 5	.....	.....	76	−10	.....	.....	76	1
12	1	2	352	−39	.....	.....	340	−31	.....	.....	107	1
13	1	1	92	−10	.....	.....	98	− 5	.....	.....	4314	11
14	1	2	88	−48	.....	.....	65	−44	.....	.....	189	1
15	1	2	90	−71	.....	.....	128	−65	.....	.....	560	18
16	1	2	63	−19	.....	.....	119	−32	.....	.....	1362	8
Savura Volcanic Group												
2	6	13	167	−61	0	34	193	−76	0	29	1256	30
Singatoka Sedimentary Group[1]												
26	2	4	351	− 2	3	15	359	−26	2	11	717	15
27	2	4	62	−23	37	53	65	−36	15	32	144	20
28	1	2	12	−44	.....	.....	19	−17	.....	.....	240	9
29	3	6	62	− 9	8	41	49	+ 5	6	37	129	16
30	3	6	2	−30	0	19	3	−38	0	12	172	8

Tholo Plutonics												
33	4	8	57	−51	1	33	341	−57	32	66	35	7
34	1	2	351	+27	.....	.....	76	−43	.....	.....	42	3
35	1	2	169	+ 2	.....	.....	145	−11	.....	.....	1037	29
41[5]	4	7	232	+77	22	64	46	+58	67	79	118	17
Wainimala Group[2]												
6	1	1	158	−29	.....	.....	97	−15	.....	.....	165	5
7	1	1	51	+17	.....	.....	44	+30	.....	.....	58	3
8	3	4	242	+43	82	86	60	− 6	63	77	67	8
9	2	4	9	−13	41	57	356	−26	15	32	198	23
10	2	4	77	−36	1	10	73	−25	1	7	99	3
11	2	3	347	−39	71	81	30	−44	1	8	385	35
17	2	4	151	+52	42	57	117	− 1	30	47	521	31
18	3	6	346	−48	5	37	7	−64	6	38	338	22
20	2	4	62	+31	13	30	59	+15	1	6	300	16
21	1	2	109	−17	.....	.....	.....	.....	.....	.....	1	1
22	2	4	107	+40	57	70	111	+10	79	89	113	17
23	2	4	74	− 9	69	79	84	−33	24	42	308	40
24	1	1	76	−34	.....	.....	329	+56	.....	.....	373	106
25	1	1	46	−64	.....	.....	342	+ 7	.....	.....	238	33
31	1	2	302	−85	.....	.....	145	+ 4	.....	.....	37	8
32	2	4	16	− 5	64	75	23	−24	33	50	502	36
36	1	2	304	+ 7	.....	.....	300	+ 3	.....	.....	5699	37
37	1	1	.....	.....	.....	.....	.....	.....	.....	.....	1	1
38	1	1	.....	.....	.....	.....	.....	.....	.....	.....	1	1
39	2	3	357	− 1	83	93	41	−59	36	52	156	3
40[6]	1	2	33	−24	.....	.....	57	−16	.....	.....	275	8
42	1	2	328	+28	.....	.....	262	+28	.....	.....	357	70

[1]All sedimentary rocks, with variable degrees of weathering.
[2]All sedimentary rocks often slightly metamorphosed or weathered (except for igneous sample at site 24).
[3]Age of one sample 4·3 ± 0·2 m.y. (N. H. Gale, pers. comm.).
[4]Lomolomo quarry; age of other samples from this quarry 5·35 ± 0·1 m.y. (McDougall, 1963).
[5]Age of one sample 33 m.y. (McDougall, 1963).
[6]Limestone containing upper Eocene fossils.

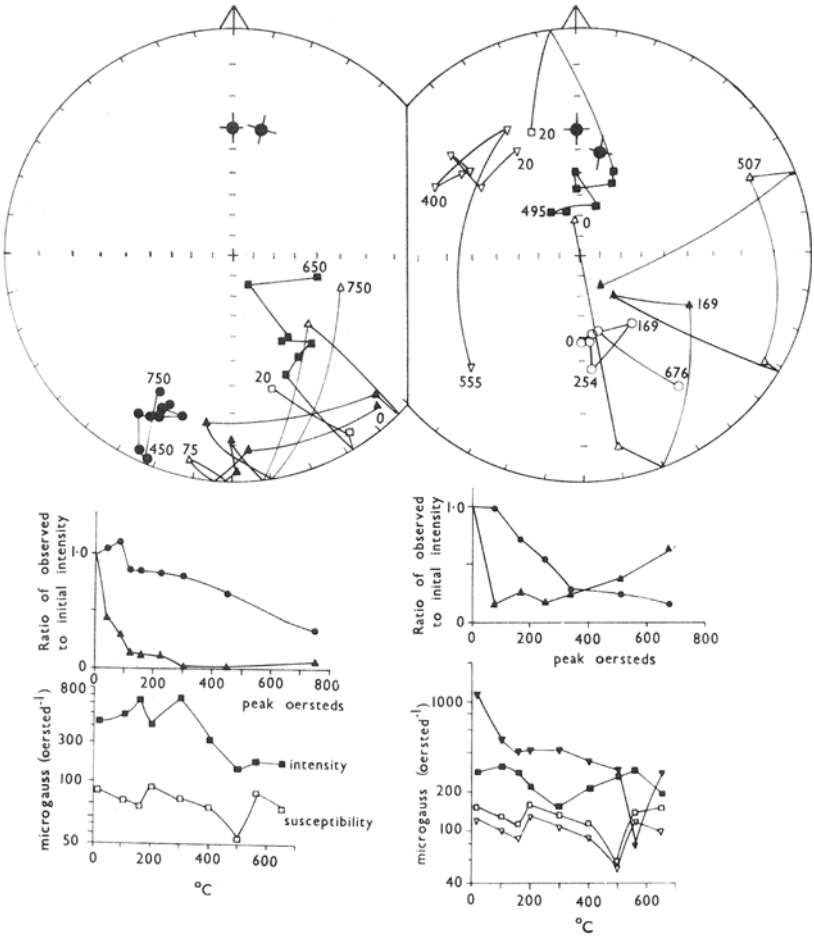


FIG. 4—Partial demagnetisation of pilot specimens from the Mba Basaltic Group. (See also Table 2.) Solid and hollow symbols as in Fig 2. Axial dipole and Earth's magnetic field direction also shown.

Left-hand Figures

□ 124 (alternating mag. field)
△ 124 (thermal)
○ 123 (alternating mag. field)
(all from site 57)

Right-hand Figures

inverted triangle 111 from site 51 (thermal)
□ 116 from site 53 (thermal)
△ 101 from site 47 (a.c.)
○ 107 from site 49 (a.c.)

TABLE 2—Results of Partial Demagnetisation of Pilot Specimens

Site	Specimen	D (°)	I (°)	Most Stable Range: Thermal (°C)	Most Stable Range: Alternating Magnetic Field (oersted)	Stability Index
Mba Basaltic Group						
51	111	308	-24	160-300	-	2.4
53	116	356	+64	200-650	-	1.8
57	124	146	+39	200-500	-	2.6
47	101a	110	+34	-	169-676	0.5
49	107a	170	-48	-	0-507	2.3
57	123a	204	+13	-	0-750	3.6
57	124b	182	+ 5	-	37-150	1.0
Mendrausuthu Andesitic Group (Marls)						
15	6a	322	-17	-	37-150	4.2
16	20a	69	- 3	-	75-150	1.0
Singatoka Sedimentary Group						
26	51a	346	-26	-	37-150	2.9
26	52a	7	-29	-	75-150	1.7
Tholo Plutonics						
41	85a	105	+62	-	75-150	2.7
41	86a	244	+54	-	0-750	1.3
Wainimala Group						
17	38a	144	-73	-	225-750	1.1
17	39b	144	- 3	-	225-750	1.5

D, I are the declination and inclination of the remanent vector over the most stable range.

Stability Index (Tarling and Symons, 1967) is given by the maximum value of (range $\frac{1}{2}$/circular standard deviation) for three or more successive observations of the remanent vector after successive increases in temperature or applied alternating field. The range selected by this formula is regarded as the most stable range. A stability index of 2·5 or greater is thought to indicate good stability in most cases.

(x) *Thuvu Sedimentary Group* comprises local deposits of calcareous sandstones and beach deposits, which lie on an erosion surface of probably Pliocene age. One site was sampled (Table 1). The mean remanent direction was similar before and after treatment in 150 (peak) oersteds, but the individual specimen directions became strongly divergent after treatment.

(xi) *Vanua Levu.* Two samples were obtained by Dr M. J. Rickard from this island. One sample (A) was from a basaltic flow of the Natewa Basic-Andesite Volcanic Group on Ndelaikoro Mountain, and the other (B) was from one of the Vutusinga Plugs of the Naroro Hornblende Andesite Group at Ndelanathau. Both had remanent directions which were southerly and of low inclination, but agreement between the two sample directions was poor.

CONCLUSIONS

The paucity of sampling severely restricts the reliability of any conclusions attempted on this data. However, it is clear that almost all the sedimentary rocks sampled are unstable or poorly stable. In many cases, this instability may be associated with weathering effects, but it is felt that most sediments in this region are basically unsuitable for paleomagnetic analysis. The volcanic rocks appear to be generally moderately stable, although the directions determined often appear to be anomalous in the older volcanics, i.e., the directions are often neither similar to nor reversed with respect to the Earth's present magnetic field or the axial geocentric dipole field. The Mba Basaltic Group, which was sampled in most detail, contains some distinct polarity zones which may be related to reversals of the Earth's magnetic field. The Mba samples are of varying stability and any subsequent paleomagnetic investigation would have to involve detailed collection and laboratory analysis.

Thorough paleomagnetic investigation of the volcanic sequences in Viti Levu and neighbouring islands could be expected to yield results useful in the elucidation of the age relationships of certain lava sequences and of the tectonic history of this region, which is of fundamental importance in deciphering the geological history of the South-West Pacific. Unfortunately, the results presented here cannot be considered sufficiently reliable for this type of interpretation.

ACKNOWLEDGMENTS

The sampling and most of the measurements were undertaken while the author held a Research Scholarship in the Department of Geophysics and Geochemistry of the Australian National University, and subsequent measurements and analysis were completed in the Department of Geophysics and Planetary Physics of the University of Newcastle upon Tyne. The author is indebted to both institutions and their staffs for financial and academic support. The sampling in Fiji was completed with the assistance of members of the Fiji Geological Survey, in particular R. E. Houtz, M. J. Rickard, and T. Naisua. The geology of Viti Levu has recently undergone considerable modification following detailed mapping by the Fiji Geological Survey, and the author is indebted to Dr M. J. Rickard, P. Rodda, R. P. Band, and Professor W. R. Dickinson for discussion of these developments, although the author is solely responsible for the views expressed.

REFERENCES

BLACKETT, P. M. S. 1952: A negative experiment relating to magnetism and the Earth's rotation. *Phil. Trans. Roy. Soc., Lond. A, 245*: 309–70.

COLLINSON, D. W.; CREER, K. M.; IRVING, E.; RUNCORN, S. K. 1957: Palaeomagnetic measurements in Great Britain—I. *Phil. Trans. Roy. Soc., Lond. A, 250*: 73–82.

CREER, K. M. 1958: Preliminary palaeomagnetic measurements from South America. *Ann. Geophys. 14*: 373–90.

IRVING, E.; STOTT, P. M.; WARD, M. A. 1961a: Demagnetization of igneous rocks by alternating magnetic fields. *Phil. Mag. 6*: 225–41.

IRVING, E.; ROBERTSON, W. A.; STOTT, P. M.; TARLING, D. H.; WARD, M. A. 1961b: Treatment of partially stable sedimentary rocks showing planar distribution of magnetization. *J. Geophys. Res. 66*: 1927–33.

MCDOUGALL, I. 1963: Potassium–argon ages of some rocks from Viti Levu, Fiji. *Nature, 198*: 677.

RODDA, P.; BAND, R. B. 1967: Geology of Viti Levu. (Abstract.) *In* South-west Pacific Geological Conference, Fiji, 1966. Proceedings and Abstracts. PHILLIPS, K. A. (Comp.). *N.Z. Jl Geol. Geophys. 10 (5) (this issue)*: 1179–80.

TARLING, D. H. 1963: A palaeomagnetic study of some late Cenozoic rocks in low latitudes. Ph.D. thesis, Australian National University: 1–360.

TARLING, D. H.; SYMONS, D. T. A. 1967: A stability index of remanence in paleomagnetism. *Geophys. J. Roy. Astr. Soc. 12*: 443–8.